

Assignment: Python Basics and Data Structures

Part 1: Python Introduction and Data Types

1. Personal Information

Write a Python program that stores and prints the following information about yourself:

- Name
- Age
- Height
- Whether you are a student (True or False)

Use appropriate Python data types for each value.

Example output:

Name: John

Age: 25

Height: 1.75

Student: True

2. Identify the Data Types

Given the following variables:

```
name = "Abdurrahman"
```

```
age = 25
```

```
height = 1.75
```

```
is_student = True
```

Write a program that prints the **value and data type** of each variable.

Expected concept:

```
print(type(name))
```

Part 2: Lists

3. Favourite Foods

Create a list containing at least **5 favourite foods**.

Perform the following operations:

1. Print the entire list.
 2. Print the first and last food.
 3. Add one more food.
 4. Remove one food.
 5. Change one food to another food.
 6. Print the final list.
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4. Student Scores

Given the following list:

scores = [75, 80, 65, 90, 85]

Write a program to:

1. Print all the scores.
 2. Print the highest score.
 3. Print the lowest score.
 4. Add a new score of 95.
 5. Print the updated list.
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Part 3: Tuples

5. Days of the Week

Create a tuple containing the seven days of the week.

Your program should:

1. Print the entire tuple.
2. Print the first day.
3. Print the last day.
4. Attempt to change one of the values and explain what happens.

Question: Why is a tuple different from a list?

Part 4: Sets

6. Remove Duplicate Values

Given the following list:

```
numbers = [1, 2, 3, 4, 2, 5, 3, 6, 1]
```

Write a program to:

1. Convert the list into a set.
 2. Print the result.
 3. Explain why some values disappeared.
-

7. Unique Programming Languages

Given:

Languages: Python, Java, Python, C++, JavaScript, Python

Convert the list into a set and print the unique programming languages.

Then add "Django" to the set.

Part 5: Dictionaries

8. Student Profile

Create a dictionary containing information about a student:

Name

Age

Course

Level

Skills

Perform the following:

1. Print the entire dictionary.
 2. Print the student's name.
 3. Add a new key called email.
 4. Change the student's level.
 5. Remove the age key.
 6. Print the final dictionary.
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Final Challenge

9. Student Management Data

Create a dictionary for **3 students**. Each student should have:

- Name
- Age
- Course
- A list of at least 3 skills

Example structure:

1. student1:
 - a. name: John
 - b. age: 22
 - c. course: Backend Development,
 - d. skills: Python, HTML, Git
2. student2:
 - a. name: Mary
 - b. age: 24
 - c. course: Data Analysis
 - d. skills: Excel, SQL, Python

Your program should print the information for all students.

Submission Requirement

Students should submit:

1. A Python file named assignment1.py.
2. Their code should be properly commented.
3. Each question should be clearly labelled in the code.
4. Screenshots of the program output.

Assignment: Student Information Manager

Objective

Create a Python program that stores and displays information about a student using different Python data types and data structures.

Instructions

Create variables containing the following student information:

1. **Name** – String

2. **Age** – Integer
3. **Height** – Float
4. **Is currently enrolled** – Boolean

Then create the following data structures:

1. List

Create a list containing at least **5 programming languages or subjects** the student is interested in.

Example:

Skills: Python, HTML, CSS, JavaScript, SQL

Perform the following:

- Print the first item.
 - Add a new item.
 - Remove one item.
 - Print the updated list.
-

2. Tuple

Create a tuple containing the student's **3 favorite numbers**.

Example:

Favorite numbers = 7, 10, 25

Print the second number.

3. Set

Create a set containing some hobbies. Include at least one duplicate value.

Example:

Hobbies: Reading, Gaming, Football, Reading

- Print the set.
 - Explain through the program output what happened to the duplicate value.
 - Add a new hobby.
-

4. Dictionary

Create a dictionary containing all the student's basic information.

Example structure:

name: John

age: 25

height: 1.75,

is enrolled: True

skills: Python, SQL, Excel

favorite numbers: 7, 10, 25

hobbies: Reading, Gaming, Football

Then:

- Print the student's name.
 - Print their skills.
 - Add a new key called "country".
 - Update the student's age.
 - Print the complete dictionary.
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★ Bonus Challenge

Create a variable to store your:

- Name
- Age
- Favorite programming language

Then store the information in a dictionary and print something like:

Hello Abdurrahman!

You are 25 years old.

Your favorite programming language is Python.

Submission

submit:

1. Their .py file.
2. A screenshot showing the program running successfully.

3. A short explanation of the difference between:

- List
- Tuple
- Set
- Dictionary